

Canonicity for free categories follows a structure like,

$$\begin{array}{ccc}
 & \xrightarrow{\exists s} & \text{Set}^D \\
 F_Q & \xrightarrow{F_Q(1_r)} & \text{Set} \\
 & \downarrow & \\
 & & \text{Set}
 \end{array}$$

where F_Q is the free category over a quiver Q ,

$F_Q(1, \cdot)$ is the global sections functor sending a type to closed terms of that type, and Set^D can be shown to have the necessary (displayed) universal structure for invoking the universal property of F_Q

For instance, the free cartesian cartesian closed bicartesian closed category eliminates into any displayed category w/ appropriate structure

The idea is that you interpret each generator of the quiver into a type that has canonical forms.

For instance booleans are true or false.

If the quiver is given by

$$\begin{array}{ccc}
 1 & \xrightarrow{z} & \mathbb{N}^e \\
 t \downarrow & & \downarrow f \\
 \text{Bool} & & \text{Set}
 \end{array}$$

Then we interpret each base type into the obvious set

We want to invoke the local eliminator of F_Q and the Bicartesian^v of Set^D to get a section

$$\begin{array}{ccc}
 & \xrightarrow{\quad} & \text{Set}^D \\
 F_Q & \xrightarrow{\quad} & \text{Set} \\
 & \downarrow & \\
 & & \text{Set}
 \end{array}$$

With this section we can interpret each base type into its displayed set of canonical forms.

That is base type A gets interpreted as the displayed set

$$\text{isCanonical}_A : \llbracket A \rrbracket_{\text{Set}} \rightarrow \text{hSet}$$

And each morphism generator must map to a displayed morphism in Set^D

That is, $\llbracket \text{Bool} \rrbracket_{\text{Set}} : \llbracket \text{Bool} \rrbracket_{\text{Set}} \rightarrow \text{hSet}$
 $\llbracket \text{Bool} \rrbracket_{\text{Set}} e := e \equiv t + e \equiv f$
 syntactically

And $\llbracket \mathbb{N} \rrbracket_{\text{Set}} : \llbracket \mathbb{N} \rrbracket_{\text{Set}} \rightarrow \text{hSet}$
 $\llbracket \mathbb{N} \rrbracket_{\text{Set}} e := e \equiv z + \sum_{n : \llbracket \mathbb{N} \rrbracket_{\text{Set}}} (\llbracket \mathbb{N} \rrbracket_{\text{Set}} n) \wedge (e \equiv h * s)$

Then,

$$\llbracket t \rrbracket_{\text{Set}} : \text{Set}^D[\llbracket t \rrbracket_{\text{Set}}][\llbracket 1 \rrbracket_{\text{Set}} \llbracket \text{Bool} \rrbracket_{\text{Set}}]$$

Max has it set up so that we build canonical forms for $\text{id} * e$ rather than e directly. Why?

The canonicalization section

$$\begin{array}{ccc}
 & \xrightarrow{\text{canon}} & \text{Set}^D \\
 F_Q & \xrightarrow{\text{pts}} & \text{Set} \\
 & \downarrow & \\
 & & \text{Set}
 \end{array}$$

when acting a term, i.e. a morphism, must

use the action of the section on morphisms

So we must use $\text{canon} \cdot F \text{hom}^D e$ which is a function from closed terms of unit type to the type of e ?